AGNEL CHARITIES

FR. C. RODRIGUES INSTITUTE OF TECHNOLOGY,VASHI

DEPARTMENT OF ELECTRICAL ENGINEERING

IN ASSOCIATION WITH IEEE FCRIT STB 61121, EESA & ISLE FCRIT

PRESENTS

OVERCLOCK 2027

A 24-HOUR NATIONAL-LEVEL HARDWARE HACKATHON

29TH -30TH JAN 2027

PRIZE POOL WORTH ₹75,000

IDEAS
CIRCUITS
PROTOTYPES
PEOPLE
PROGRESS



FR. C RODRIGUES INSTITUTE OF TECHNOLOGY
AGNEL TECHNICAL EDUCATION COMPLEX, SECTOR 9-A, VASHI, NAVI MUMBAI, MAHARASHTRA – 400703, INDIA

SCAN TO REGISTER

ROUND 1: ONLINE & FREE // ROUND 2: 24-HR ON-SITE FINALE

WHAT IS OVERCLOCK 2027 ?

OVERCLOCK 2027 is a 24-hour national hardware hackathon where engineers, makers, and innovators turn bold concepts into working prototypes. Build under real-world constraints, push the boundaries of hardware design, and engineer tangible solutions for tomorrow.

YOUR MISSION:

Step into the timeline, confront the challenge, and engineer the hardware that redefines tomorrow.

THE INCIDENT

THE THEME SECTION

What if critical technologies never got built? Across alternate timelines, major systems failed before they could work.

Your job is to step into these scenarios, tackle the engineering breakdown, and build working hardware prototypes to fix the timeline.



HANDS-ON INNOVATION	REAL WORLD IMPACT	LEARN FROM EXPERTS	NETWORK WITH LIKE-MINDED MAKERS

DOMAINS

EXPLORE BUILD STABILIZE

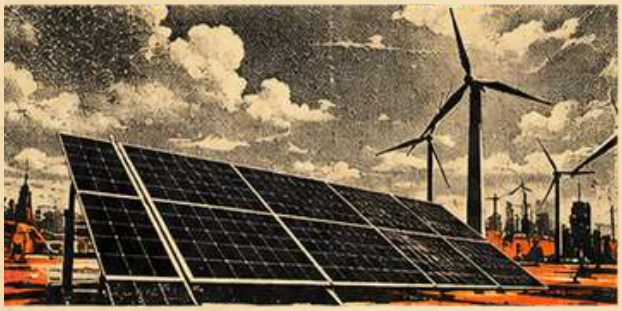
GENERAL INNOVATION & OPEN INNOVATION

Everyday problems, accessibility, smart solutions.



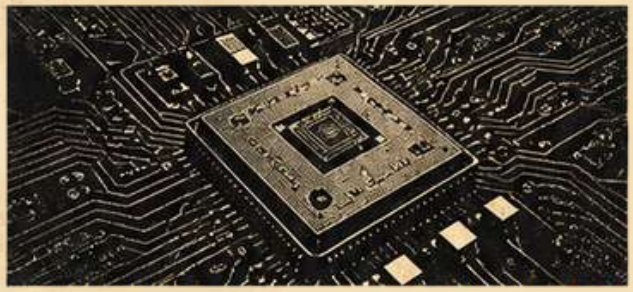
ENERGY, POWER & SUSTAINABILITY

Renewables, EVs, smart grids, power quality, energy management.



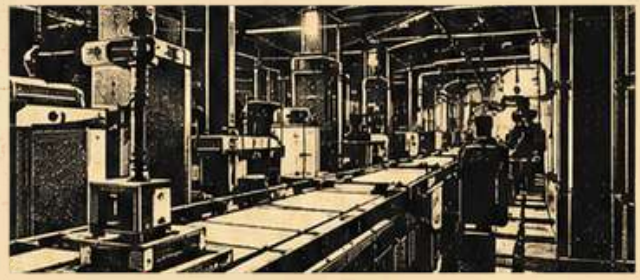
SMART HOME & CONSUMER ELECTRONICS

Appliance automation, energy efficiency, safety



INDUSTRIAL AUTOMATION & MONITORING .

Predictive maintenance, fault detection, machine monitoring



DEFENCE & NATIONAL SECURITY

Surveillance, threat detection, secure communication, rescue systems.



HEALTHCARE & ASSISTIVE TECHNOLOGY

Patient monitoring, rehabilitation, emergency response

IDEAS
CIRCUITS
PROTOTYPES
PEOPLE

BUILD
ITERATE
BRANCH
REPEAT

REAL-WORLD
IMPACT

A BRIGHTER
TOMORROW

THE TIMELINE

FROM IDEA TO IMPACT.

6TH NOV — **REGISTRATIONS OPEN**
Free online registration begins for Round 1

10TH NOV — **PROBLEM STATEMENT RELEASE**
Round 1 challenge statements live on website & Unstop

1-15TH DEC — **PPT SUBMISSION**
Submit technical approach & proposed solution template

22ND DEC — **SELECTED TEAMS ANNOUNCEMENT**
Declaration of top 30 qualified + 5 waitlisted teams

22ND JAN — **FINAL PROBLEM STATEMENT**
Fresh statements released for component sourcing & planning

29TH JAN — **WELCOME TO OVERCLOCK 2027**
Reporting & kickoff of the 24-hr on-site hardware build at FCRIT

30TH JAN — **SUBMISSION, JUDGING & WINNER ANNOUNCED**
Live demonstrations, final evaluation & prize distribution



RULES & REGULATIONS

- TEAMS OF 2-4 MEMBERS
- ELIGIBILITY UG STUDENTS
- AI USAGE / ASSISTIVE ONLY
- HARDWARE / BYOC POLICY
- ORIGINAL WORK / ZERO PLAGIARISM
- LAB SAFETY / PROTOCOLS
- LIVE PROTOTYPING / NO PRE-BUILT KITS
- JUDGING / JURY DECISION FINAL

AVAILABLE RESOURCES



ELECTRICAL MACHINES & TESTING FACILITIES

1. DC MOTOR TESTING FACILITY
2. THREE-PHASE INDUCTION MOTOR TESTING FACILITY
3. PSP / TRANSMISSION LINE SIMULATOR
4. THREE-PHASE 415 V AC SUPPLY
5. THREE-PHASE AUTOTRANSFORMER
6. SINGLE-PHASE VARIAC

ELECTRICAL & ELECTRONIC TEST EQUIPMENT

1. DIGITAL MULTIMETERS
2. DIGITAL OSCILLOSCOPES
3. DC POWER SUPPLIES
4. DUAL-OUTPUT ±12 V POWER SUPPLIES
5. FUNCTION GENERATORS

TOOLS & WORKSHOP FACILITIES

1. SOLDERING IRONS
2. SOLDERING IRON STANDS
3. EXTENSION BOARDS
4. BASIC ELECTRONIC AND ELECTRICAL TOOLS
5. PCB PRINTING / FABRICATION SERVICE
6. 3D PRINTING SERVICE

NOTE:

LAB FACILITIES AND STANDARD TEST GEAR ARE ACCESSIBLE SUBJECT TO LAB SAFETY PROTOCOLS.

HEAVY/SPECIALIZED LAB GEAR IS LIMITED AND ALLOCATED ON A STRICT FIRST-COME, FIRST-SERVED (FCFS) BASIS.

COLLEGE TEST BENCHES & BENCHMARK RIGS MAY ONLY BE OPERATED UNDER LAB IN-CHARGE SUPERVISION.

SPECIAL TOOL OR COMPONENT REQUIREMENTS MUST BE REQUESTED IN ADVANCE TO THE ORGANIZING COMMITTEE.

JUDGING CRITERIA

EVALUATED BY INDUSTRY & ACADEMIC EXPERTS

IDEATION & PROBLEM CONTEXT	10%
INNOVATION & CREATIVITY	20%
WORKING FUNCTIONALITY	15%
RELEVANCE & REAL-WORLD IMPACT	15%
FEASIBILITY & SCALABILITY	10%
HARDWARE INTEGRATION & RELIABILITY	20%
PRESENTATION& Q&A	10%



PRIZES & REWARDS

RECOGNIZING TIMELINE CHANGERS.

FIRST RUNNER-UP (2ND PLACE)
₹ 15,000

GRAND PRIZE (1ST PLACE)
₹ 25,000

SECOND RUNNER-UP (3RD PLACE)
₹ 10,000

CONSOLATION & CATEGORY PRIZES: ₹25,000

EVERY PARTICIPANT GETS

- PARTICIPATION CERTIFICATE
- MEALS & REFRESHMENTS
- DEDICATED WORKSPACE & WI-FI
- EXPERT MENTORSHIP
- PARTICIP ANT KIT & SWAG



VENUE & REGISTRATION

SAME IDEAS.
DIFFERENT
TIMELINES.

KEY DATES & EVENT DETAILS

FINALE DATES
29TH-30TH JANUARY

VENUE
FCRIT CAMPUS, VASHI, NAVI MUMBAI, MAHARASHTRA

REGISTRATION OPENS
6 NOV 2026

SUBMISSIONS CLOSE (ROUND 1)
15TH DECEMBER 2026

REGISTRATION FEE
ROUND 1: FREE
ROUND 2 (SHORTLISTED ONLY): ₹1,500 + 18% GST / TEAM



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IEEE STUDENT BRANCH FCRIT

FOLLOW US
FOR UPDATES

ROUND 1 - ONLINE PROPOSAL

ENTRY FEE: FREE (NO REGISTRATION OR PARTICIPATION CHARGES)

GUIDELINES:

- SUBMISSION FORMAT: TEAMS MUST SUBMIT THEIR PROPOSED HARDWARE SOLUTION STRICTLY USING THE OFFICIAL PRESENTATION TEMPLATE PROVIDED ON UNSTOP AND THE WEBSITE.
- REQUIRED SECTIONS: THE PROPOSAL MUST COVER PROBLEM UNDERSTANDING, PROPOSED TECHNICAL APPROACH, WORKING PRINCIPLE, PROPOSED HARDWARE COMPONENTS, TECHNICAL FEASIBILITY, AND EXPECTED IMPACT.
- EVALUATION METRICS: SUBMISSIONS WILL BE EVALUATED BY AN EXPERT PANEL BASED ON INNOVATION, TECHNICAL DEPTH, FEASIBILITY, AND REAL-WORLD IMPLEMENTATION POTENTIAL.
- QUALIFICATION: TOP-SCORING TEAMS WILL BE SHORTLISTED TO ADVANCE DIRECTLY TO THE ROUND 2 ON-SITE 24-HOUR FINALE.
- KEY DEADLINES: SUBMISSIONS OPEN FROM 1ST DECEMBER 2026 AND CLOSE STRICTLY ON 15TH DECEMBER 2026. THE SHORTLIST WILL BE DECLARED ON 22ND DECEMBER 2026.

NOTE

- ORIGINALITY: SUBMISSIONS MUST BE THE TEAM'S ORIGINAL CONCEPTUAL WORK; COPIED OR PLAGIARIZED IDEAS WILL BE DISQUALIFIED IMMEDIATELY.
- OFFICIAL TEMPLATE: DEVIATING FROM THE REQUIRED TEMPLATE SLIDE COUNT OR FORMAT MAY LEAD TO PENALTY POINTS DURING SCREENING.
- NO HARDWARE NEEDED IN ROUND 1: ROUND 1 IS STRICTLY AN ONLINE PROPOSAL EVALUATION. NO PHYSICAL PROTOTYPE DEMONSTRATION IS REQUIRED UNTIL ROUND 2.

ROUND 2 – 24-HOUR OFFLINE FINALE

REGISTRATION FEE: ₹1,500 + 18% GST PER TEAM

GUIDELINES:

- **ELIGIBILITY & CONFIRMATION:** ONLY SHORTLISTED TEAMS FROM ROUND 1 QUALIFY TO PARTICIPATE UPON PAYING THE REGISTRATION FEE WITHIN THE NOTIFIED SLOT WINDOW.
- **FRESH PROBLEM STATEMENT:** A NEW DOMAIN PROBLEM STATEMENT WILL BE RELEASED 1 WEEK PRIOR TO THE FINALE (TENTATIVELY 22ND JANUARY 2027) STRICTLY FOR COMPONENT PROCUREMENT, CIRCUIT PLANNING, AND SIMULATIONS.
- **LIVE ON-SITE PROTOTYPING:** ALL CIRCUIT FABRICATION, BREADBOARDING, SOLDERING, ASSEMBLY, AND CODING MUST BE EXECUTED LIVE ON-SITE DURING THE CONTINUOUS 24-HOUR BUILD WINDOW AT FCRIT. PRE-ASSEMBLED CIRCUITS ARE STRICTLY BARRED.
- **FUNCTIONAL PROTOTYPE MANDATE**: TEAMS MUST PRESENT AND DEMONSTRATE AN ACTIVE, WORKING HARDWARE PROTOTYPE WITH SUPPORTING FIRMWARE/SOFTWARE DURING FINAL JURY EVALUATION.
- **PRESENTATION & DEFENSE:** TEAMS WILL DEMONSTRATE LIVE FUNCTIONALITY, EXPLAIN THE HARDWARE ARCHITECTURE, AND ANSWER QUESTIONS DURING THE TECHNICAL Q&A.
- **MEALS & ESSENTIALS INCLUDED:** REGISTRATION FEE COVERS PARTICIPANT KITS, CONTINUOUS REFRESHMENTS, MIDNIGHT SNACKS, AND ALL MEALS ACROSS DAY 1 AND DAY 2.
- **REST ZONE (INCLUDED IN FEE):** ON THE NIGHT OF 29TH JANUARY, A DESIGNATED COMMON REST/SLEEPING AREA WILL BE AVAILABLE AT ZERO EXTRA COST. NOTE: NO LUGGAGE, TOOLS, OR HARDWARE EQUIPMENT ARE PERMITTED IN THIS QUIET REST ZONE.
- **HOSTEL ACCOMMODATION (OPTIONAL / 28TH JAN):** FOR OUTSTATION TEAMS ARRIVING ON 28TH JANUARY, SEPARATE HOSTEL ROOMS ARE AVAILABLE AT A NOMINAL ADDITIONAL CHARGE. AVAILABILITY IS STRICTLY LIMITED AND ALLOCATED ON A FIRST-COME, FIRST-SERVED (FCFS) BASIS.
- **JURY DECISION:** THE EVALUATION AND SCORING DECISIONS OF THE JUDGING PANEL AND ORGANIZING COMMITTEE ARE FINAL AND BINDING.

SECTION 1: HARDWARE DEVELOPMENT GUIDELINES

- **PERMITTED COMPONENTS**: DISCRETE PASSIVE/ACTIVE COMPONENTS, SENSORS, DEVELOPMENT BOARDS (ARDUINO, ESP32, STM32, RASPBERRY PI, ETC.), ICS, AND STANDALONE BREAKOUT MODULES ARE FULLY PERMITTED.
- **STRICT NO PRE-BUILT POLICY**: PRE-ASSEMBLED SYSTEMS, SOLDERED FINAL CIRCUITS, OR COMPLETED PROTOTYPES BROUGHT FROM OUTSIDE ARE STRICTLY PROHIBITED. BREADBOARDING, SOLDERING, WIRING, AND CODE IMPLEMENTATION MUST OCCUR LIVE ON-SITE.
- **OPEN-SOURCE & AI TOOLS**: OPEN-SOURCE CODE LIBRARIES, FRAMEWORKS, REFERENCE SCHEMATICS, AND ASSISTIVE AI ARE PERMITTED. ALL THIRD-PARTY IP MUST BE CLEARLY CITED; ENTIRE COPIED TURNKEY PROJECTS WILL BE DISQUALIFIED.
- **SPECIALIZED LAB EQUIPMENT & HIGH VOLTAGE**: TEAMS REQUIRING SPECIALIZED LAB MACHINERY (E.G., AUTO-TRANSFORMERS/VARIACS, 3-PHASE AC MAINS, INDUCTION MACHINES, HEAVY POWER SUPPLIES) MUST REGISTER EQUIPMENT REQUESTS IN ADVANCE. ACCESS IS ALLOCATED STRICTLY ON A FIRST-COME, FIRST-SERVED (FCFS) BASIS AND REQUIRES LAB IN-CHARGE CLEARANCE.

SECTION 2: SAFETY & LAB PROTOCOLS

- **HIGH VOLTAGE & CHEMICAL HAZARDS:** TESTING CIRCUITS EXCEEDING 50V AC/DC, HIGH-CURRENT LOADS, OPEN FLAMES, OR HAZARDOUS CHEMICAL ETCHING REQUIRES MANDATORY PRIOR INSPECTION AND EXPLICIT SUPERVISOR APPROVAL BEFORE ENERGIZING.
- **LAB EQUIPMENT INTEGRITY:** INSTITUTE TEST BENCHES, OSCILLOSCOPES, METERS, AND TOOLS MUST NOT BE MOVED, ALTERED, OR MISUSED. ANY EQUIPMENT FAULT, BURNOUT, OR THERMAL HAZARD MUST BE REPORTED IMMEDIATELY.
- **CODE OF CONDUCT & ZERO-TOLERANCE:** DELIBERATE EQUIPMENT SABOTAGE, HARDWARE TAMPERING OF RIVAL TEAMS, ACADEMIC PLAGIARISM, HARASSMENT, OR NON-COMPLIANCE WITH OVERNIGHT CAMPUS SECURITY RULES WILL LEAD TO IMMEDIATE DISQUALIFICATION AND CAMPUS DEBARMENT.
- **SUPERVISION MANDATE:** ORGANIZER AND FACULTY COORDINATOR SAFETY INSTRUCTIONS ARE FINAL AND LEGALLY BINDING THROUGHOUT THE CONTINUOUS 24-HOUR HACKATHON.

SECTION 3: JUDGING AND EVALUATION

- **CRITERIA COMPLIANCE:** ALL TEAMS WILL BE SCORED STRICTLY IN ACCORDANCE WITH THE PUBLISHED RUBRIC WEIGHTAGES (100% TOTAL).
- **LIVE WORKING DEMONSTRATION:** PHYSICAL PROTOTYPES MUST BE FULLY OPERATIONAL AND READY FOR LIVE DEMONSTRATION AT YOUR ASSIGNED EVALUATION BOOTH WHEN THE JUDGING PANEL ARRIVES.
- **TECHNICAL VIVA & DEFENSE:** JUDGES WILL THOROUGHLY QUESTION TEAMS ON CIRCUIT TOPOLOGY, COMPONENT CHOICE, CODE EXECUTION, TROUBLESHOOTING HISTORY, AND COMMERCIAL VIABILITY.
- **INTEGRITY & PLAGIARISM CHECK:** MISREPRESENTATION OF PROGRESS, CONCEALING PRE-BUILT HARDWARE, OR FALSIFYING SENSOR DATA WILL RESULT IN IMMEDIATE DISQUALIFICATION.
- **FINALITY OF RESULTS:** THE DECISIONS AND SCORING MARKS OF THE JURY PANEL ARE CONCLUSIVE, BINDING, AND NOT SUBJECT TO APPEAL.

SECTION 4: ACCOMMODATION & VENUE GUIDELINES

- **FREE NIGHT REST ZONE (29TH JAN):** A COMMON, QUIET REST ZONE IS PROVIDED FREE OF COST INSIDE THE CAMPUS ON THE NIGHT OF 29TH JANUARY FOR SHORT NAPS. TO MAINTAIN SPACE AND QUIET, PERSONAL LUGGAGE, TOOLKITS, AND HARDWARE EQUIPMENT ARE STRICTLY BARRED FROM THIS AREA.
- **PRE-EVENT HOSTEL STAY (OPTIONAL / 28TH JAN):** OUTSTATION TEAMS ARRIVING ON 28TH JANUARY CAN REQUEST SEPARATE HOSTEL ROOMS AT NOMINAL CHARGES. SLOTS ARE STRICTLY LIMITED AND CONFIRMED PURELY ON A FIRST-COME, FIRST-SERVED (FCFS) BASIS.
- **SECURITY & BELONGINGS:** WHILE GENERAL CAMPUS SECURITY IS MAINTAINED AROUND THE CLOCK, TEAMS ARE SOLELY RESPONSIBLE FOR THEIR PERSONAL BELONGINGS, LAPTOPS, AND SPECIALIZED HARDWARE KITS.
- **VENUE CONDUCT & PROTOCOLS:** ALL PARTICIPANTS MUST STRICTLY ADHERE TO CAMPUS MOVEMENT BOUNDARIES, LAB SAFETY DIRECTIVES, AND OVERNIGHT VENUE ACCESS RULES.
- **SCHEDULE DISCRETION:** THE ORGANIZING COMMITTEE RESERVES THE RIGHT TO MAKE OPERATIONAL SCHEDULE MODIFICATIONS IF TECHNICAL OR LOGISTICAL CONTINGENCIES ARISE.
- **RULE ACCEPTANCE:** REGISTRATION AND PARTICIPATION IN OVERCLOCK 2027 CONSTITUTE FULL AND BINDING ACCEPTANCE OF ALL STATED TERMS AND GUIDELINES.

THANK YOU

LOOKING FORWARD TO HOSTING INDIA'S BRIGHTEST HARDWARE MINDS AT FCRIT VASHI!

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REGISTRATION LINK:

HTTPS://UNSTOP.COM/O/OWM1JNP?LB=USEMSKX5&UTM_MEDIUM=SHARE&UTM_SOURCE=JAISWRAK2105&UTM_CAMPAIGN=ONLINE_CODING_CHALLENGE

WEBSITE LINK:

HHTTPS://OVERCLOCK-FCRIT.VERCEL.APP/

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